

Set 21

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. Import the dataset **in R**. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.

A. CODE:

```
data <- data.frame(
  Sector = c("Residential","Commercial","Industrial",
            "Residential","Commercial","Industrial"),
  Region = c("North","South","West","East","North","South"),
  Month = c("Jan","Jan","Feb","Feb","Mar","Mar"),
  Temperature = c(15,24,20,18,28,30),
  Units_Consumed = c(320,540,880,350,610,920),
  Cost = c(2100,3600,5900,2300,4100,6200),
  Renewable_Usage = c(22,18,12,25,20,15),
  Peak_Hours = c(4,6,8,5,7,9)
)

hist(data$Units_Consumed,
     main="Histogram of Units Consumed",
     xlab="Units Consumed (kWh)",
```

```
col="lightblue")
```

```
plot(density(data$Units_Consumed),
```

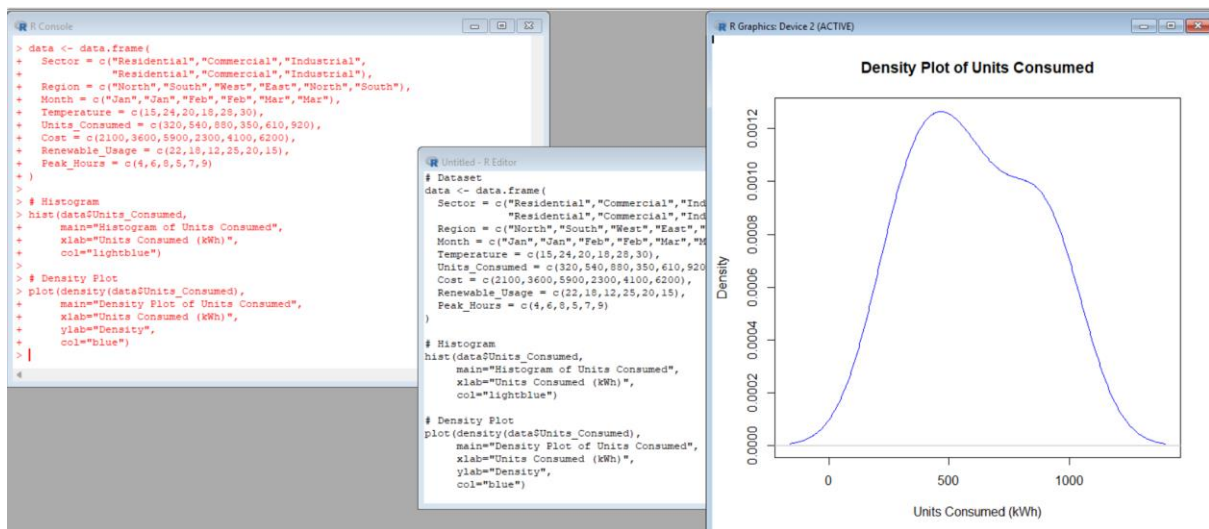
```
main="Density Plot of Units Consumed",
```

```
xlab="Units Consumed (kWh)",
```

```
ylab="Density",
```

```
col="blue")
```

OUTPUT:



2.Using **R programming**, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.

A. CODE:

```
data <- data.frame(
```

```
  Sector = c("Residential","Commercial","Industrial",
```

```
            "Residential","Commercial","Industrial"),
```

```
  Region = c("North","South","West","East","North","South"),
```

```
  Month = c("Jan","Jan","Feb","Feb","Mar","Mar"),
```

```
  Temperature = c(15,24,20,18,28,30),
```

```
  Units_Consumed = c(320,540,880,350,610,920),
```

```
  Cost = c(2100,3600,5900,2300,4100,6200),
```

```
  Renewable_Usage = c(22,18,12,25,20,15),
```

```
  Peak_Hours = c(4,6,8,5,7,9)
```

```
)
```

```
plot(data$Temperature,
```

```

data$Units_Consumed,

main="Temperature vs Units Consumed",

xlab="Temperature (C)",

ylab="Units Consumed (kWh)",

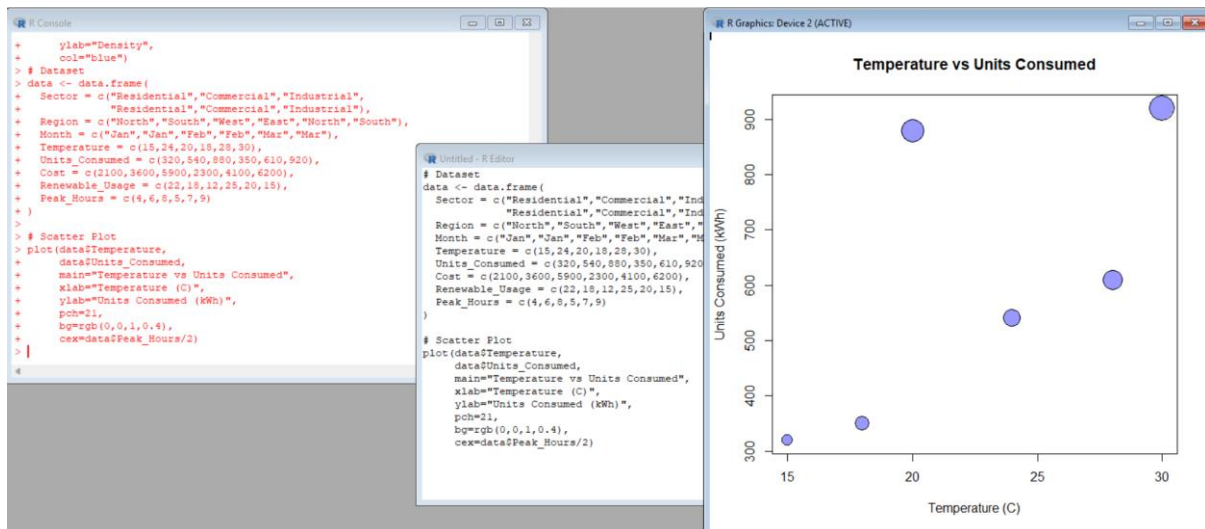
pch=21,

bg=rgb(0,0,1,0.4),

cex=data$Peak_Hours/2)

```

OUTPUT:



3. Using **R**, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.

A. CODE:

```

data <- data.frame(

  Sector = c("Residential", "Commercial", "Industrial",

             "Residential", "Commercial", "Industrial"),

  Region = c("North", "South", "West", "East", "North", "South"),

  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),

  Temperature = c(15, 24, 20, 18, 28, 30),

  Units_Consumed = c(320, 540, 880, 350, 610, 920),

  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),

  Renewable_Usage = c(22, 18, 12, 25, 20, 15),

  Peak_Hours = c(4, 6, 8, 5, 7, 9)

)

```

```

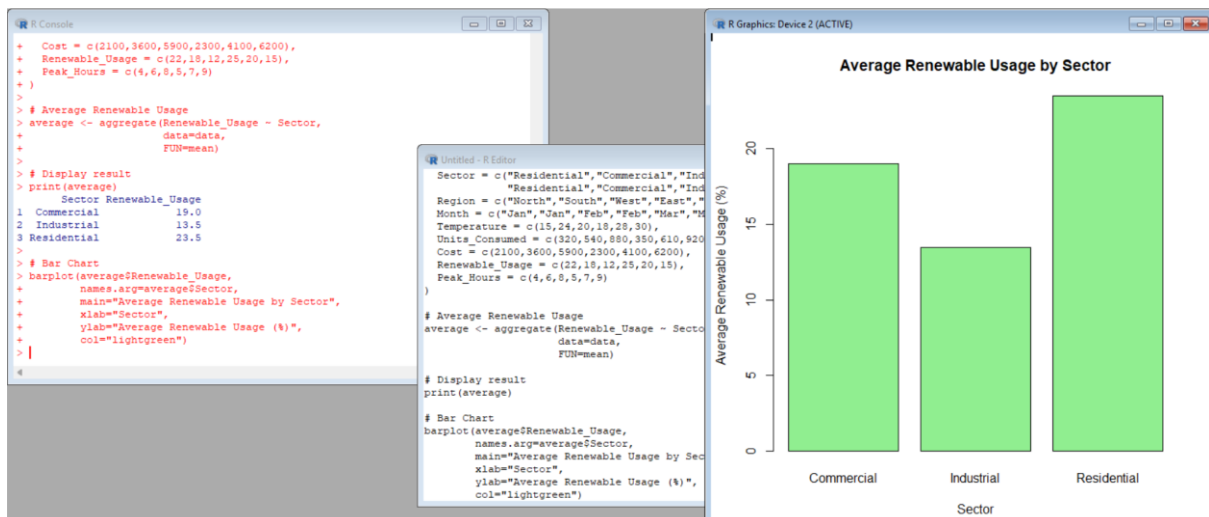
average <- aggregate(Renewable_Usage ~ Sector,
                     data=data,
                     FUN=mean)

print(average)

barplot(average$Renewable_Usage,
        names.arg=average$Sector,
        main="Average Renewable Usage by Sector",
        xlab="Sector",
        ylab="Average Renewable Usage (%)",
        col="lightgreen")

```

OUTPUT:



4.Import Energy_Consumption_Data.csv into **Tableau**. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.

A.CODE:

Q4 - Energy Consumption Dashboard

Install package once if needed

```
install.packages("plotly")
```

Load package

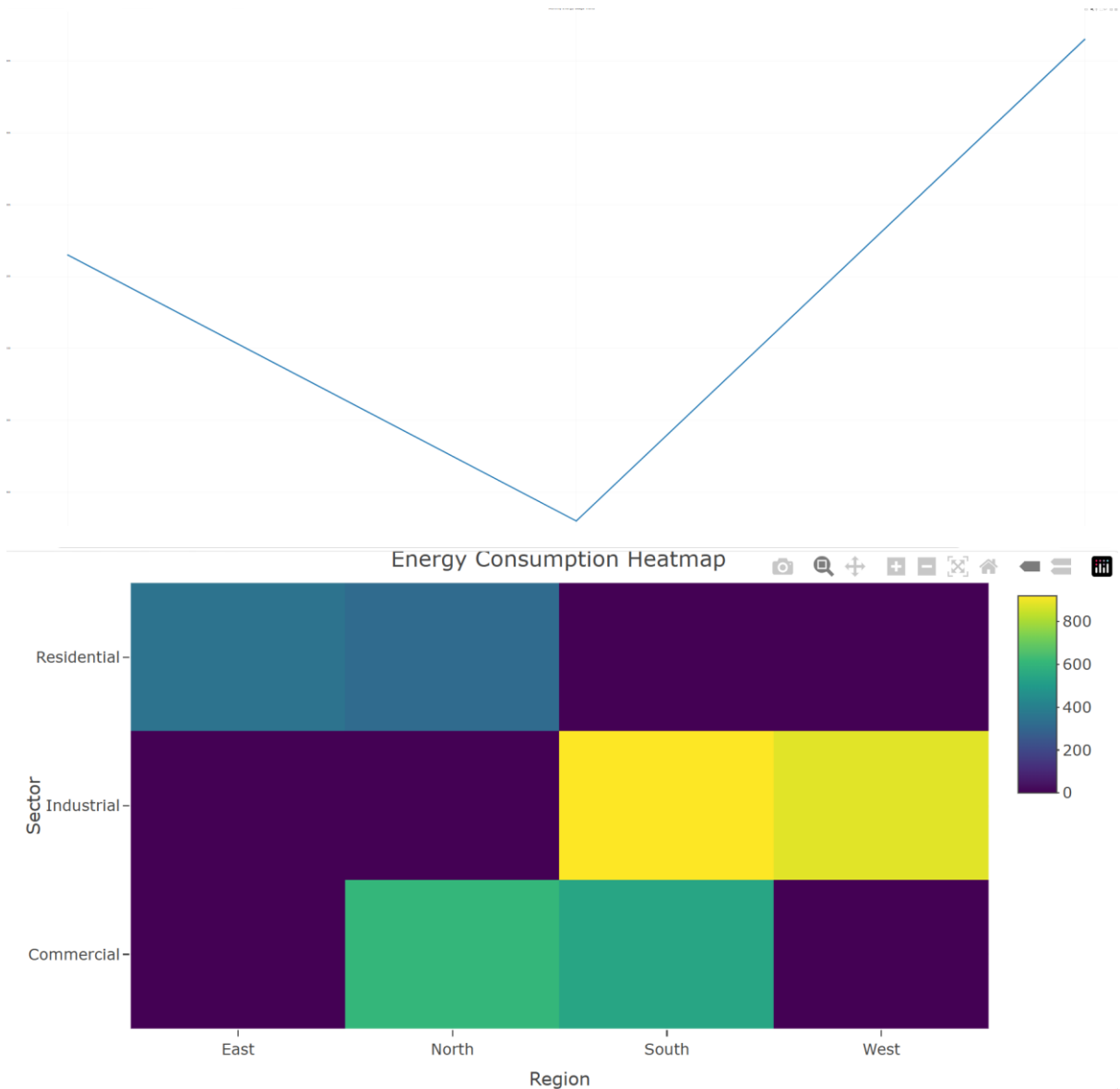
```
library(plotly)
```

```

# Dataset
data <- data.frame(
  Sector = c("Residential", "Commercial", "Industrial",
            "Residential", "Commercial", "Industrial"),
  Region = c("North", "South", "West", "East", "North", "South"),
  Month = c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar"),
  Temperature = c(15, 24, 20, 18, 28, 30),
  Units_Consumed = c(320, 540, 880, 350, 610, 920),
  Cost = c(2100, 3600, 5900, 2300, 4100, 6200),
  Renewable_Usage = c(22, 18, 12, 25, 20, 15),
  Peak_Hours = c(4, 6, 8, 5, 7, 9)
)

# Line Chart
line_chart <- plot_ly(
  monthly,
  x = ~Month,
  y = ~Units_Consumed,
  type = "scatter",
  mode = "lines+markers"
) %>%
layout(
  title = "Monthly Energy Usage Trend",
  xaxis = list(title = "Month"),
  line_chart
)
heatmap_chart

```



Set 22: Time Series Analysis

Dataset: Monthly Sales Data

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

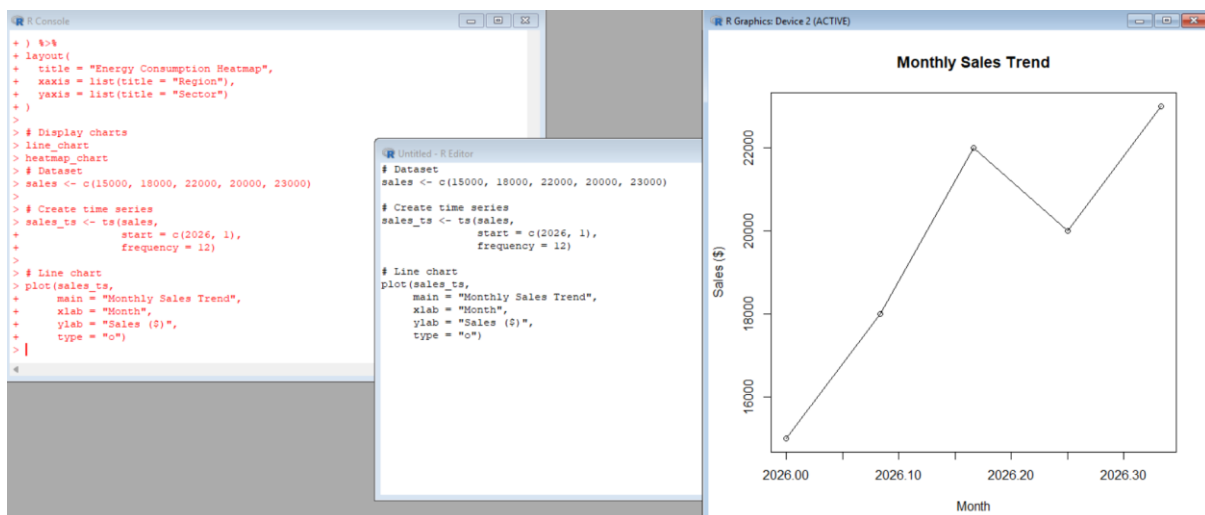
1. In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

CODE:

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
sales_ts <- ts(sales,  
               start = c(2026, 1),  
               frequency = 12)
```

```
plot(sales_ts,  
     main = "Monthly Sales Trend",  
     xlab = "Month",  
     ylab = "Sales ($)",  
     type = "o")
```



2. In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

CODE

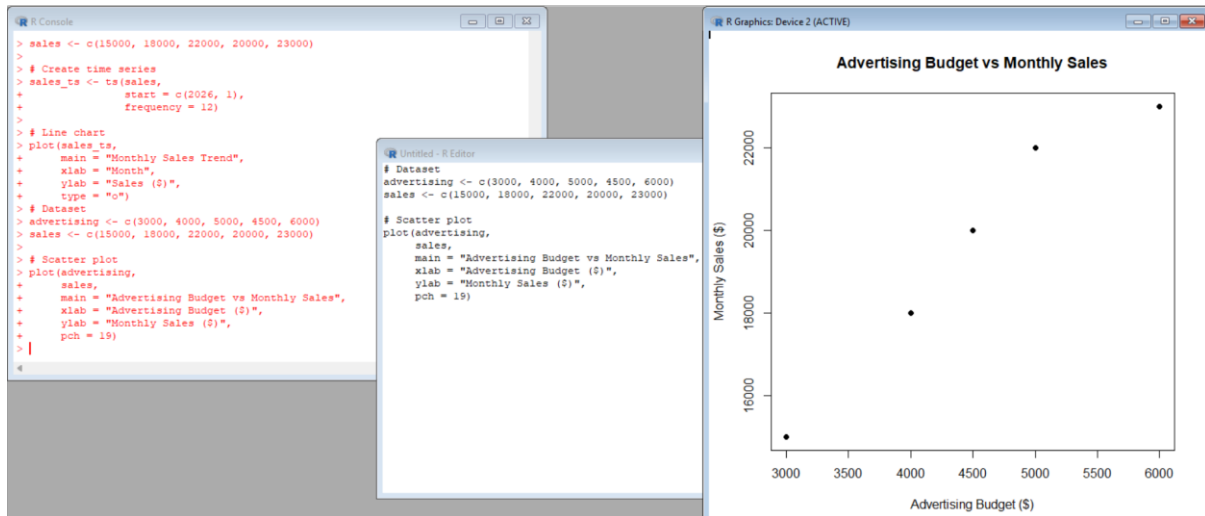
```
advertising <- c(3000, 4000, 5000, 4500, 6000)
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
plot(advertising,  
     sales,  
     main = "Advertising Budget vs Monthly Sales",  
     xlab = "Advertising Budget ($)",  
     ylab = "Monthly Sales ($)",
```

pch = 19)

OUTPUT:



3. In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

A. CODE:

```
Month <- c("January", "February", "March", "April", "May")
```

```
Sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
plot(Sales,
```

```
     type = "o",
```

```
     main = "Monthly Sales Trend",
```

```
     xlab = "Month",
```

```
     ylab = "Sales ($)",
```

```
     xaxt = "n")
```

```
axis(1,
```

```
     at = 1:5,
```

```
     labels = Month)
```

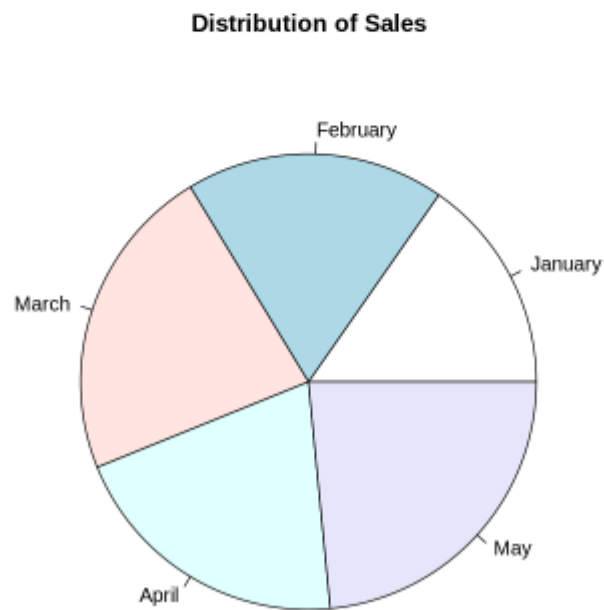
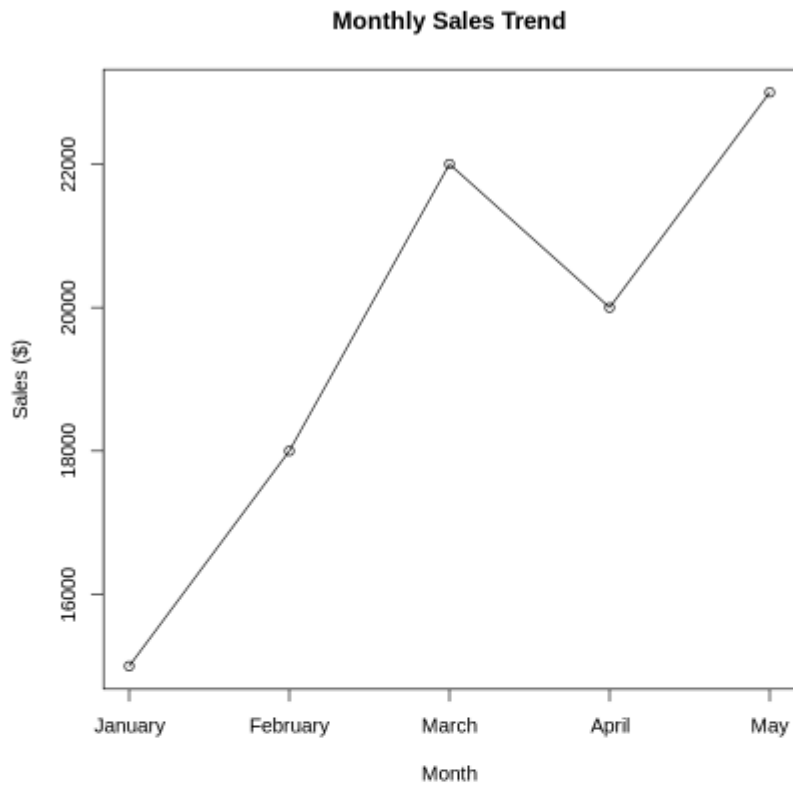
```
pie(Sales,
```

```
     labels = Month,
```

```
     main = "Distribution of Sales")
```

OUTPUT:

...



4. In R, create an autocorrelation plot to identify seasonality in the time series data.

A. CODE:

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
sales_ts <- ts(sales,  
               start = c(2026, 1),
```

```

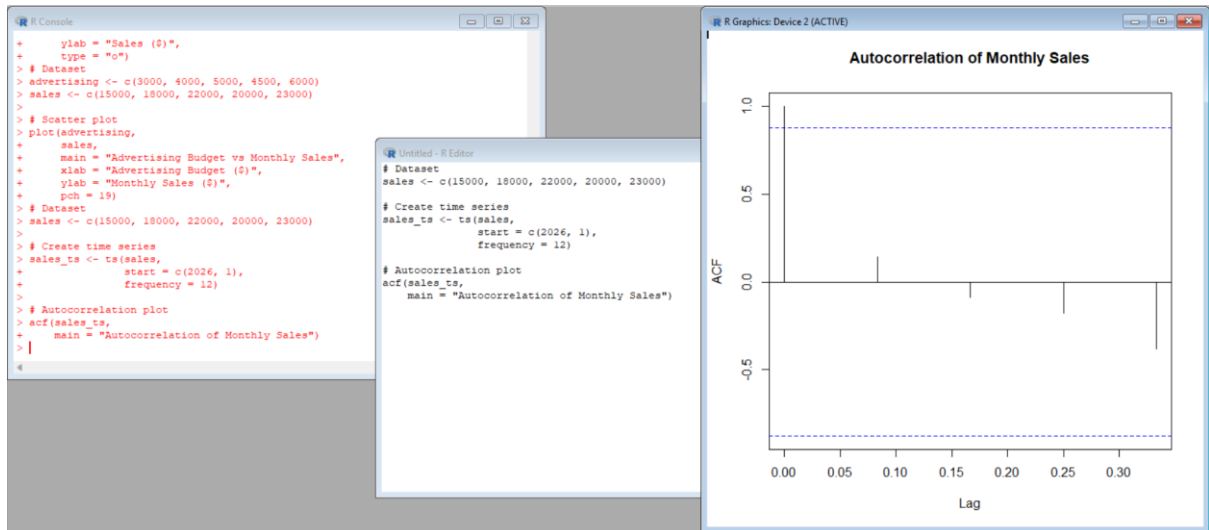
frequency = 12)

acf(sales_ts,

    main = "Autocorrelation of Monthly Sales")

```

OUTPUT:



Set 23: Employee Performance Analysis

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

CODE:

```

Employee_ID <- c(1, 2, 3, 4, 5)

Department <- c("Sales", "HR", "Marketing", "Sales", "HR")

Years_of_Service <- c(5, 3, 7, 4, 2)

Performance_Score <- c(85, 92, 78, 90, 76)

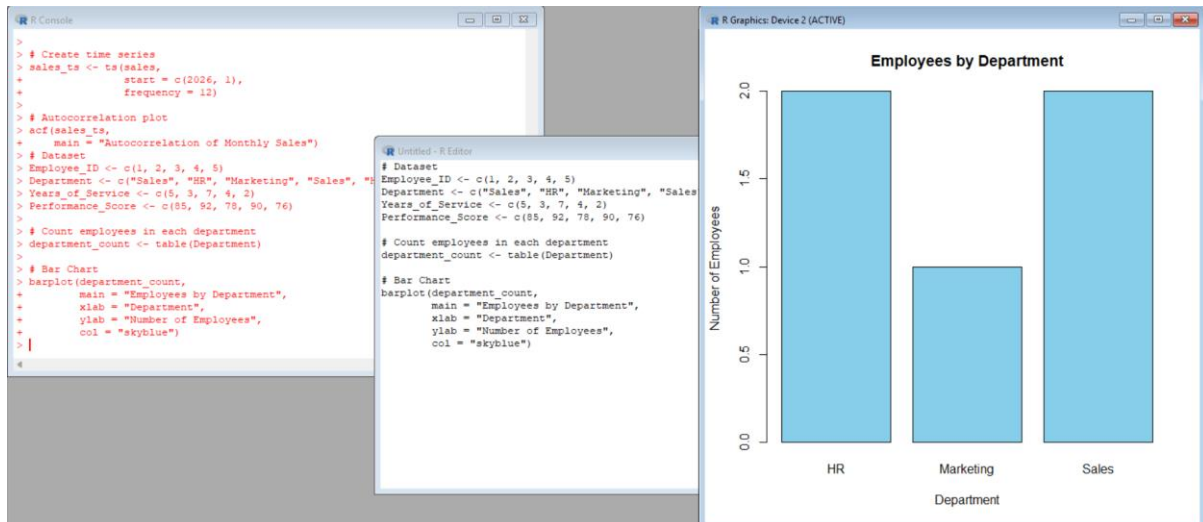
department_count <- table(Department)

```

```

barplot(department_count,
        main = "Employees by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = "skyblue")

```



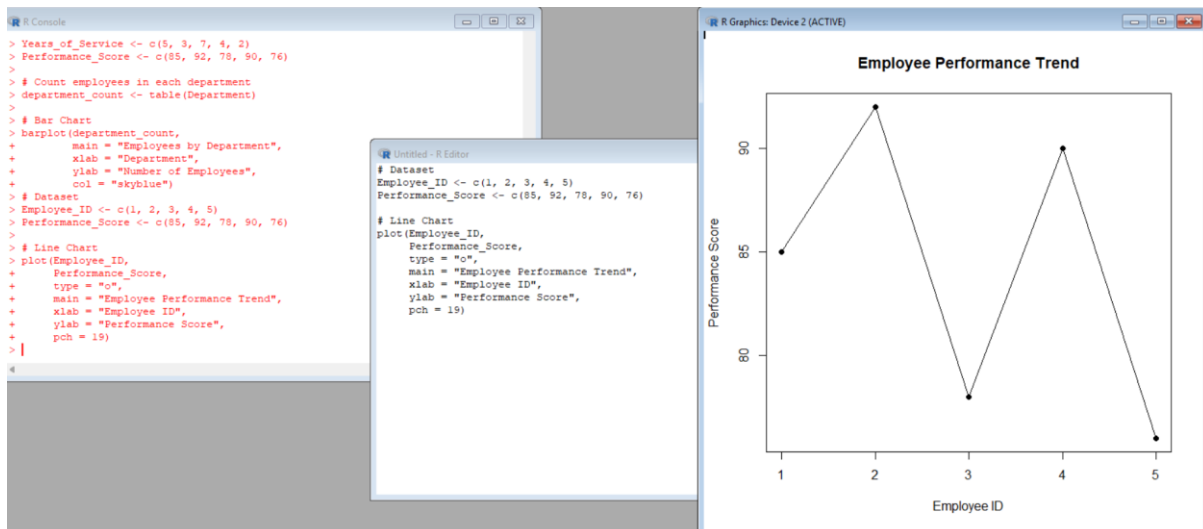
2. In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

CODE:

```

Employee_ID <- c(1, 2, 3, 4, 5)
Performance_Score <- c(85, 92, 78, 90, 76)
plot(Employee_ID,
      Performance_Score,
      type = "o",
      main = "Employee Performance Trend",
      xlab = "Employee ID",
      ylab = "Performance Score",
      pch = 19)

```



3. In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

A. CODE

```
Employee_ID <- c(1, 2, 3, 4, 5)
```

```
Department <- c("Sales", "HR", "Marketing", "Sales", "HR")
```

```
Years_of_Service <- c(5, 3, 7, 4, 2)
```

```
Performance_Score <- c(85, 92, 78, 90, 76)
```

```
department_count <- table(Department)
```

```
barplot(department_count,
```

```
  main = "Department Distribution",
```

```
  xlab = "Department",
```

```
  ylab = "Number of Employees",
```

```
  col = "skyblue")
```

```
plot(Years_of_Service,
```

```
  Performance_Score,
```

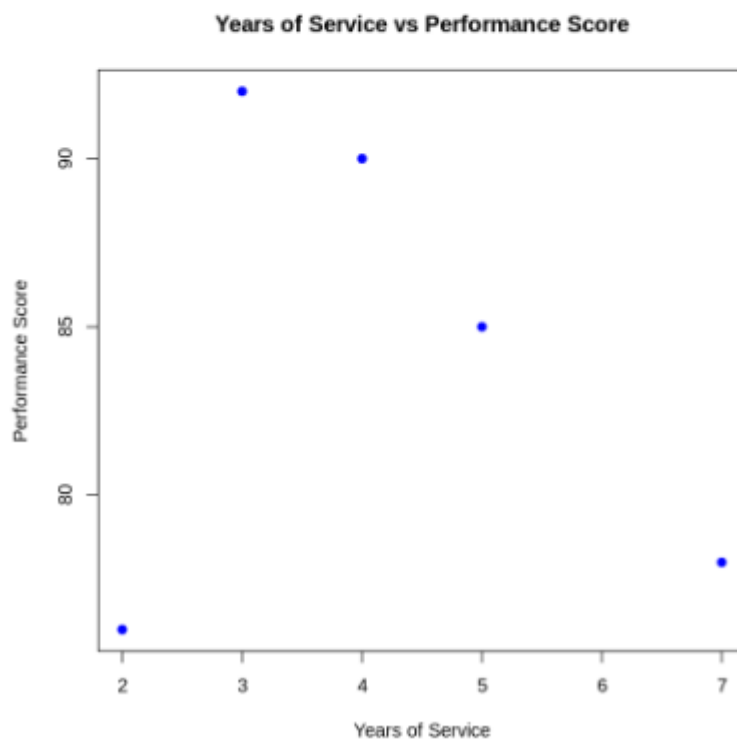
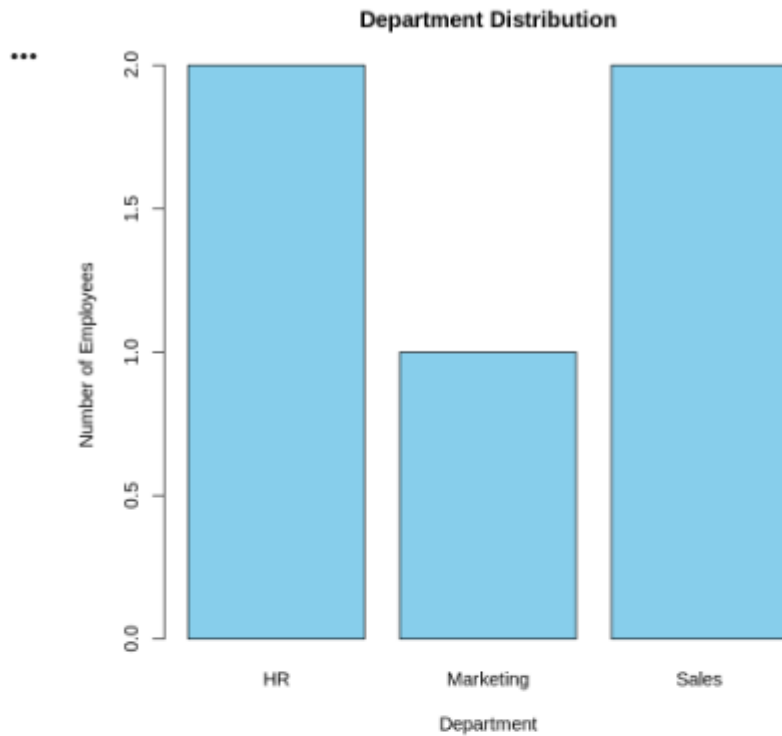
```
  main = "Years of Service vs Performance Score",
```

```
  xlab = "Years of Service",
```

```
  ylab = "Performance Score",
```

```
  pch = 19,
```

```
  col = "blue")
```



4. In R, develop a table showing the performance data for each employee. Label the table elements.

A. CODE

```
Employee_ID <- c(1, 2, 3, 4, 5)
```

```
Department <- c("Sales", "HR", "Marketing", "Sales", "HR")
```

```

Years_of_Service <- c(5, 3, 7, 4, 2)
Performance_Score <- c(85, 92, 78, 90, 76)
employee_data <- data.frame(
  Employee_ID,
  Department,
  Years_of_Service,
  Performance_Score
)
print(employee_data)

```

```

print(employee_data)

```

...	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78
4	4	Sales	4	90
5	5	HR	2	76

Set 24 Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

CODE:

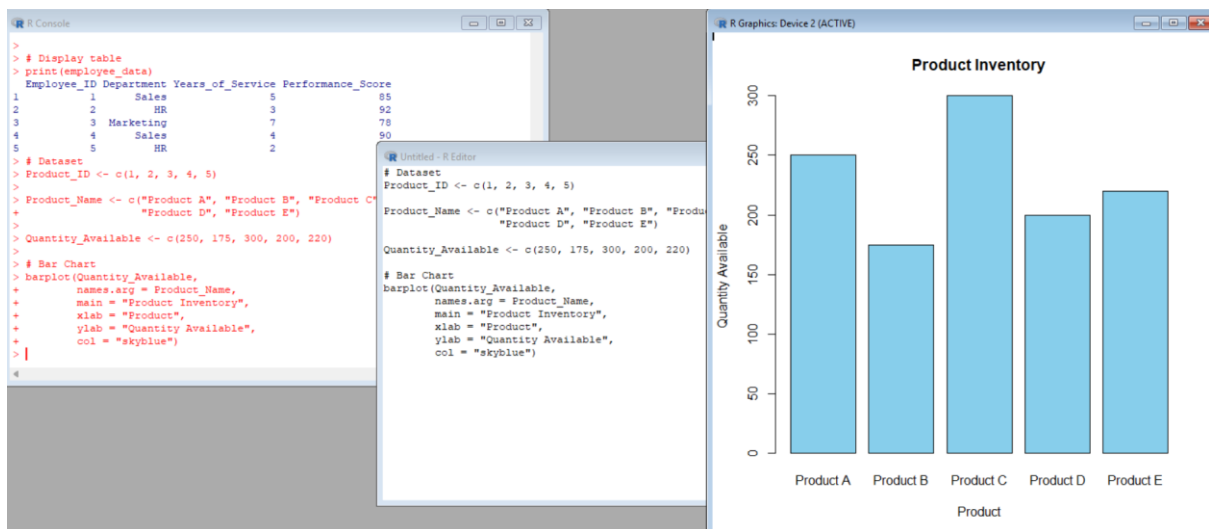
```

Product_ID <- c(1, 2, 3, 4, 5)
Product_Name <- c("Product A", "Product B", "Product C",
  "Product D", "Product E")

```

```
Quantity_Available <- c(250, 175, 300, 200, 220)
```

```
barplot(Quantity_Available,  
        names.arg = Product_Name,  
        main = "Product Inventory",  
        xlab = "Product",  
        ylab = "Quantity Available",  
        col = "skyblue")
```



2. In R, generate a stacked bar chart to show the quantity of each product within different product categories.

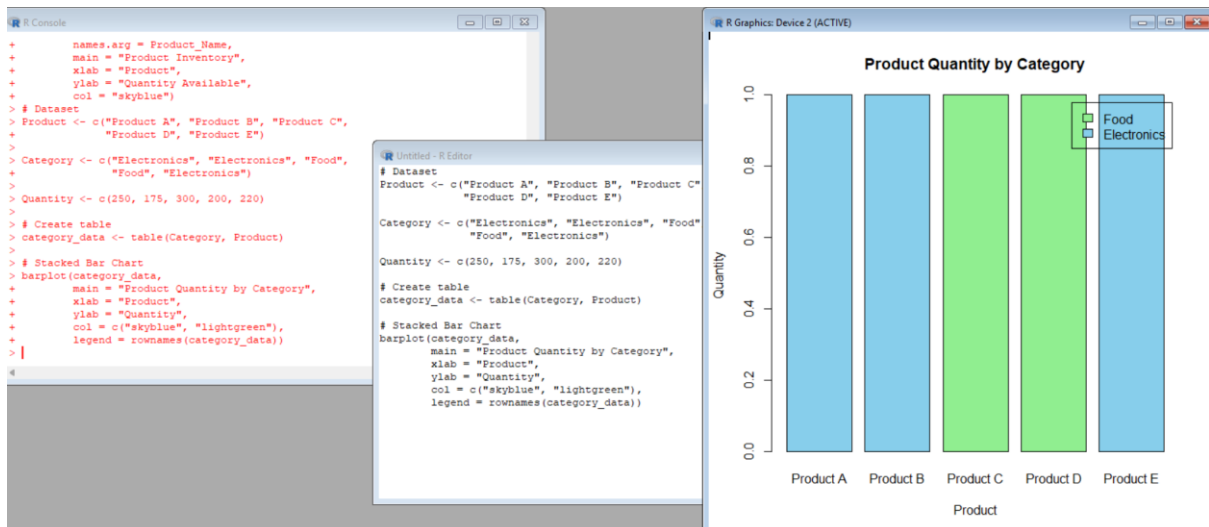
CODE

```
Product <- c("Product A", "Product B", "Product C",  
            "Product D", "Product E")
```

```
Category <- c("Electronics", "Electronics", "Food",  
            "Food", "Electronics")
```

```
Quantity <- c(250, 175, 300, 200, 220)  
category_data <- table(Category, Product)
```

```
barplot(category_data,  
        main = "Product Quantity by Category",  
        xlab = "Product",  
        ylab = "Quantity",  
        col = c("skyblue", "lightgreen"),  
        legend = rownames(category_data))
```



3. In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

CODE:

```

Product_Name <- c("Product A", "Product B", "Product C",
                  "Product D", "Product E")

```

```

Quantity_Available <- c(250, 175, 300, 200, 220)

```

```

Category <- c("Electronics", "Electronics", "Food",
              "Food", "Electronics")

```

```

barplot(Quantity_Available,
        names.arg = Product_Name,
        main = "Product Inventory",
        xlab = "Product",
        ylab = "Quantity Available",
        col = "skyblue")

```

```

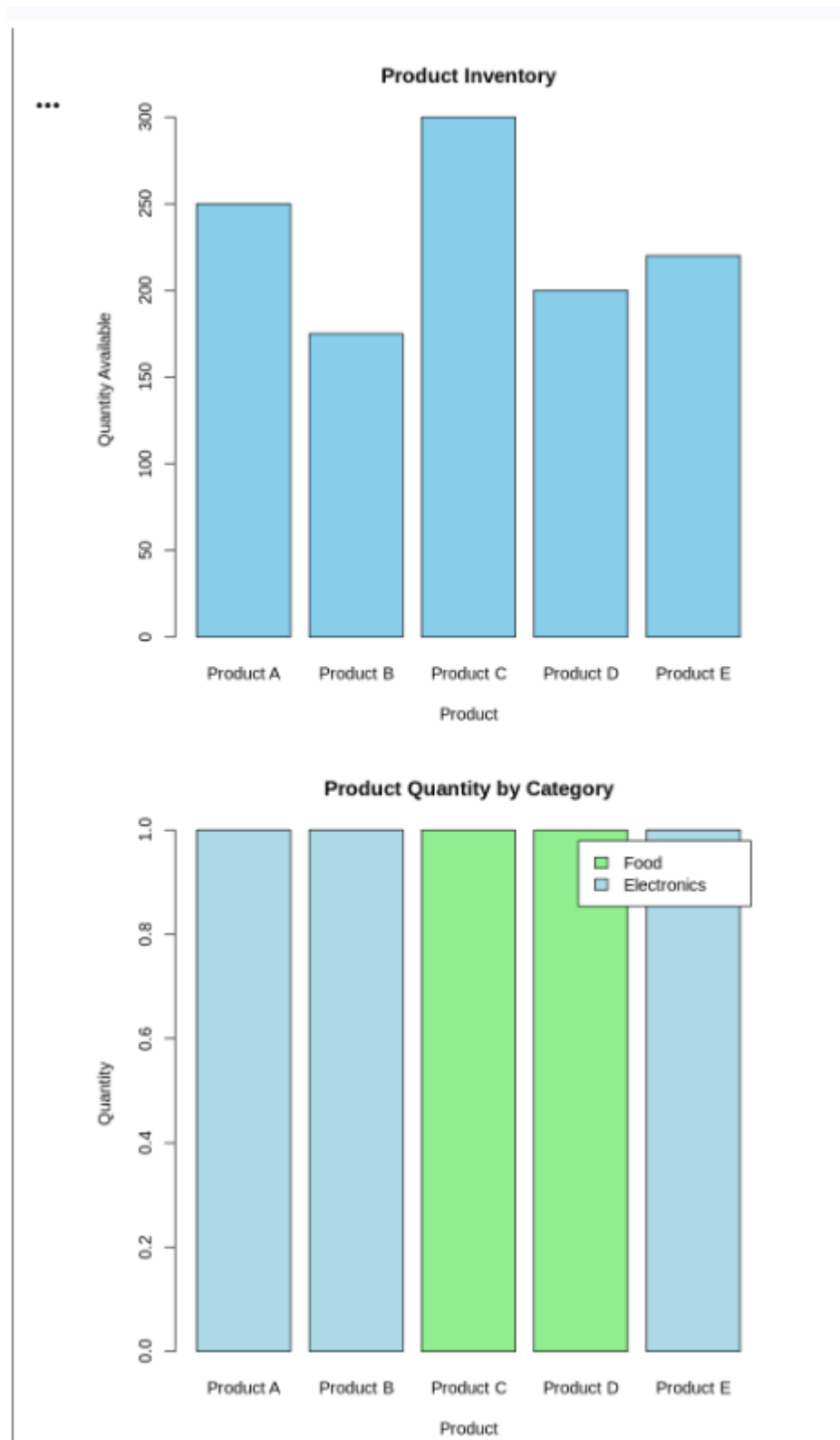
category_data <- table(Category, Product_Name)

```

```

barplot(category_data,
        main = "Product Quantity by Category",
        xlab = "Product",
        ylab = "Quantity",
        col = c("lightblue", "lightgreen"),
        legend = rownames(category_data))

```

4. In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

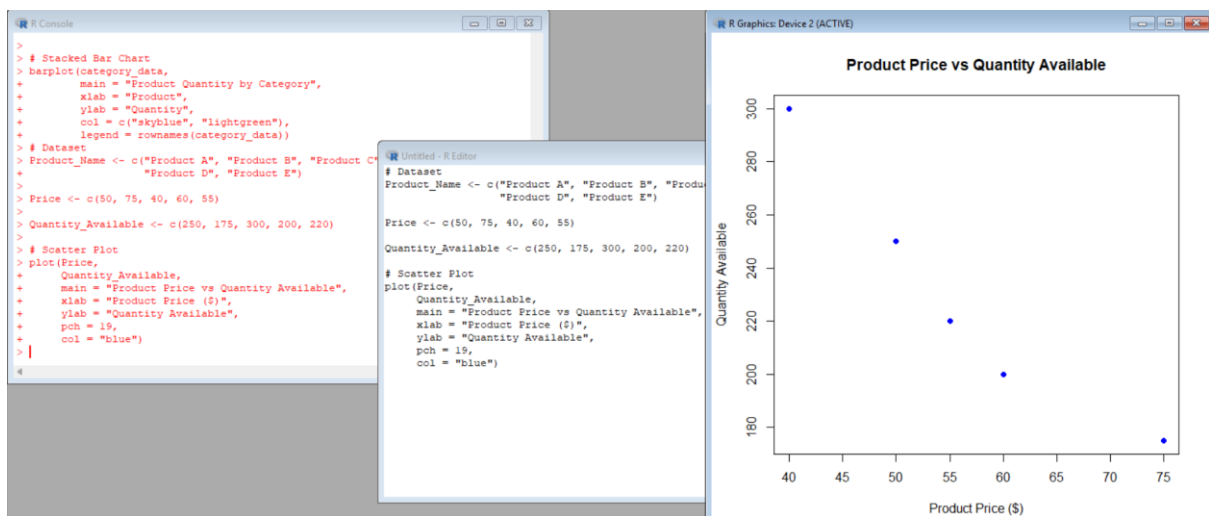
A.CODE

```
Product_Name <- c("Product A", "Product B", "Product C",
```

```

"Product D", "Product E")
Price <- c(50, 75, 40, 60, 55)
Quantity_Available <- c(250, 175, 300, 200, 220)plot(Price,
  Quantity_Available,
  main = "Product Price vs Quantity Available",
  xlab = "Product Price ($)",
  ylab = "Quantity Available",
  pch = 19,
  col = "blue")

```



Set 25: Website Traffic Analysis

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. In Tableau, create a line chart to visualize the trend in page views over time. Label the axes and title the chart.

A.CODE

```
# Dataset
```

```
Date <- c("2023-01-01", "2023-01-02", "2023-01-03",
```

```
"2023-01-04", "2023-01-05")
```

```
Page_Views <- c(1500, 1600, 1400, 1650, 1800)
```

```
Date <- as.Date(Date)
```

```
plot(Date,
```

```
Page_Views,
```

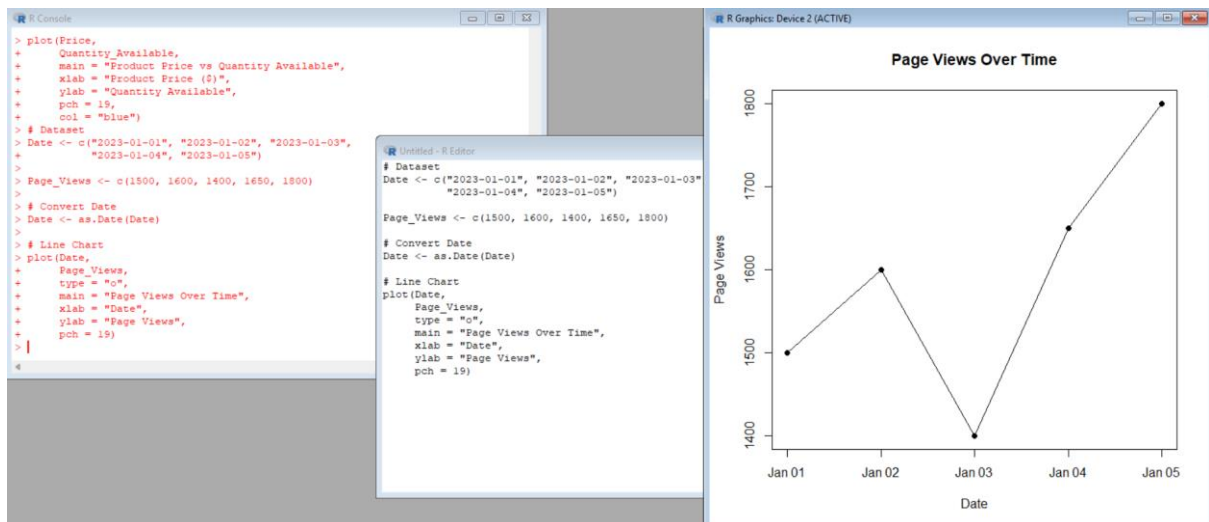
```
type = "o",
```

```
main = "Page Views Over Time",
```

```
xlab = "Date",
```

```
ylab = "Page Views",
```

```
pch = 19)
```



2. In R, generate a bar chart to show the top N days with the highest click-through rates. Label the chart elements.

CODE

```
Date <- c("2023-01-01", "2023-01-02", "2023-01-03",
```

```
"2023-01-04", "2023-01-05")
```

```
Click_Rate <- c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
top3 <- order(Click_Rate, decreasing = TRUE)[1:3]
```

```
barplot(Click_Rate[top3],
```

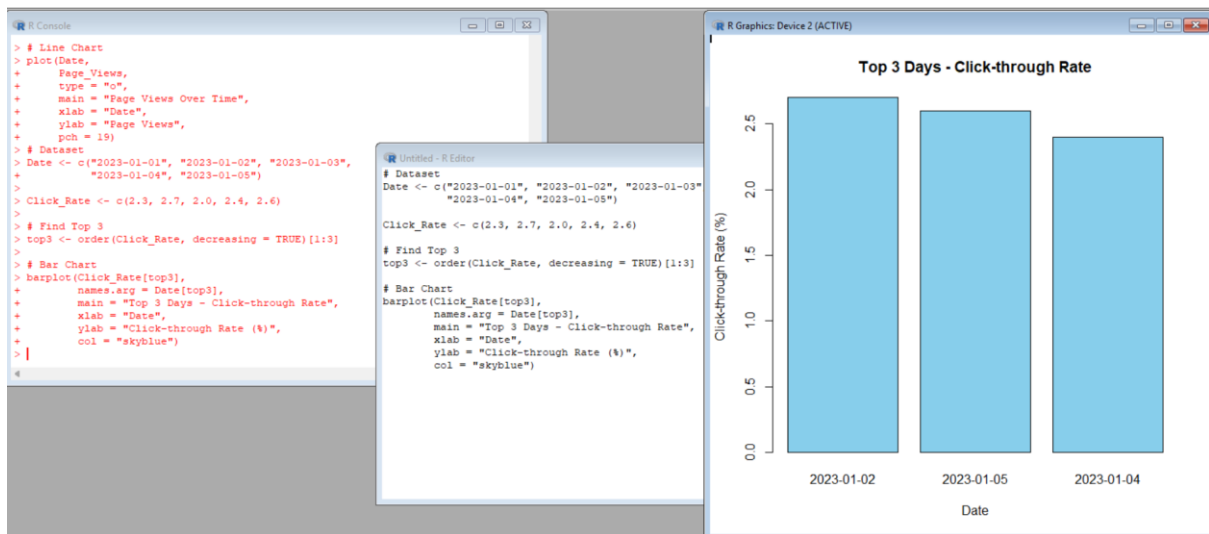
```
names.arg = Date[top3],
```

```
main = "Top 3 Days - Click-through Rate",
```

```
xlab = "Date",
```

```
ylab = "Click-through Rate (%)",
```

```
col = "skyblue")
```



3. In R, build a stacked area chart to display the distribution of user interactions (likes, shares, comments) on the website.

CODE

```
Date <- c("2023-01-01", "2023-01-02", "2023-01-03",
```

```
         "2023-01-04", "2023-01-05")
```

```
Likes <- c(300, 350, 280, 400, 450)
```

```
Shares <- c(100, 120, 90, 150, 170)
```

```
Comments <- c(50, 60, 45, 70, 80)
```

```
interaction <- rbind(Likes, Shares, Comments)
```

```
plot(Date,
```

```
     Likes + Shares + Comments,
```

```
     type = "n",
```

```
     main = "User Interactions",
```

```
     xlab = "Date",
```

```
     ylab = "Number of Interactions")
```

```
polygon(c(1:5, 5:1),
```

```
        c(Likes, rep(0,5)),
```

```
        col = "lightblue")
```

```
polygon(c(1:5, 5:1),
```

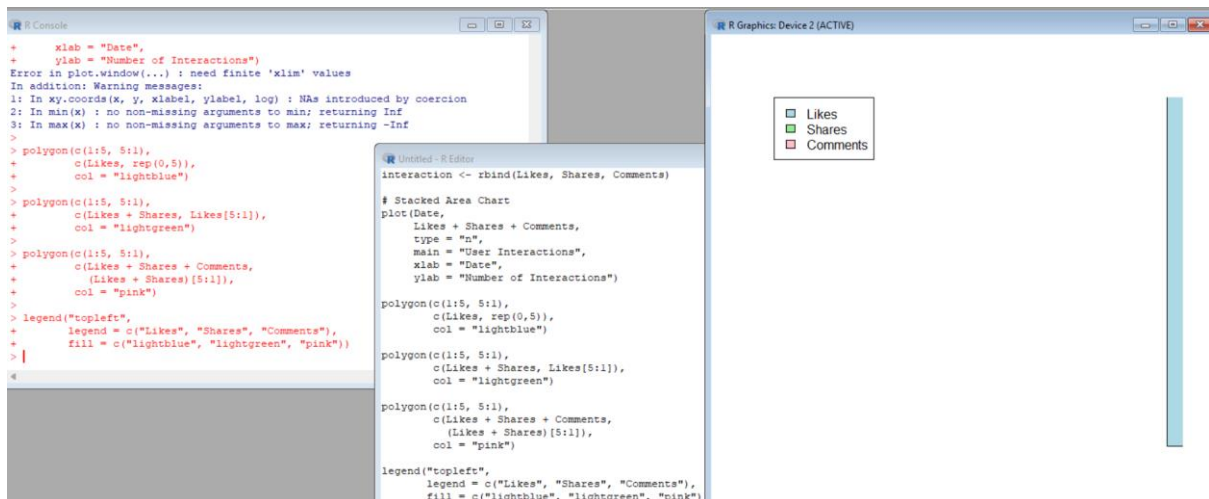
```
        c(Likes + Shares, Likes[5:1]),
```

```
        col = "lightgreen")
```

```

polygon(c(1:5, 5:1),
       c(Likes + Shares + Comments,
         (Likes + Shares)[5:1]),
       col = "pink"
legend("topleft",
      legend = c("Likes", "Shares", "Comments"),
      fill = c("lightblue", "lightgreen", "pink"))

```



4. In Tableau, create a dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

CODE

```

Source <- c("Google", "Facebook", "Instagram",
            "YouTube", "Google")

Location <- c("India", "USA", "UK",
              "Canada", "Australia")

Page_Views <- c(1500, 1200, 1000, 900, 1800)
traffic <- data.frame(
  Source,
  Location,
  Page_Views
)

print(traffic)

barplot(Page_Views,
        names.arg = Source,

```

```
main = "Page Views by Traffic Source",  
xlab = "Traffic Source",  
ylab = "Page Views",  
col = "skyblue")
```

```
***  
  Source Location Page_Views  
1  Google   India     1500  
2 Facebook   USA     1200  
3 Instagram  UK       1000  
4  YouTube  Canada     900  
5  Google Australia  1800
```

